Title

Data Cleaning and Visualization of Student Performance Dataset

I. Introduction

Data cleaning and visualization are important steps in data analysis. This project uses a student performance dataset to clean the data and visualize the results using Python libraries such as Pandas, Matplotlib, and Seaborn.

II. Objective

- To clean the dataset.
- To identify missing values and duplicate records.
- To visualize the data using graphs.
- To gain insights from the dataset.

III. Tools Used

- Google Colab
- Python
- Pandas
- Matplotlib
- Seaborn

IV. Dataset Description

The Student Performance Dataset contains information about students' scores in Mathematics, Reading, and Writing. The dataset was used to analyze student performance and create visualizations..

V. Data Cleaning process

Step 1: Loaded the dataset

The dataset was taken from Kaggle and imported into Google colab using pandas.

Step 2: Checked Missing values

The datset has been checked for any missing values or null values and if any present they can be replaced by the common or mean .

Step 3: Checked for Duplicate values

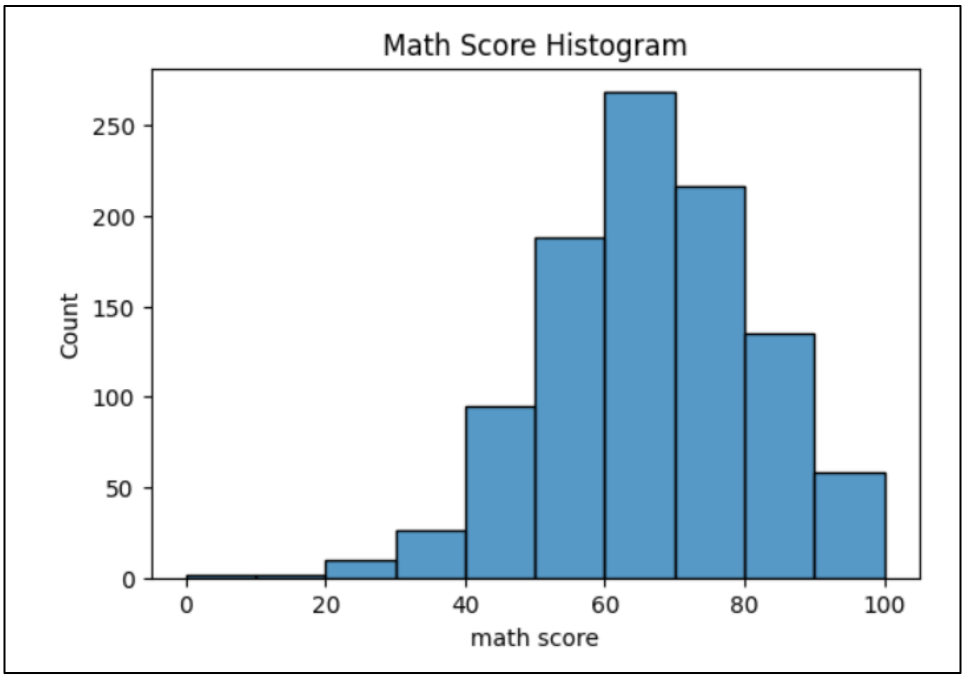
If any duplicate values or records are present then they are removed.

Step 4:Verification of Data Quality

After all the cleaning of the dataset then it was reviewed to ensure consistency in the data.

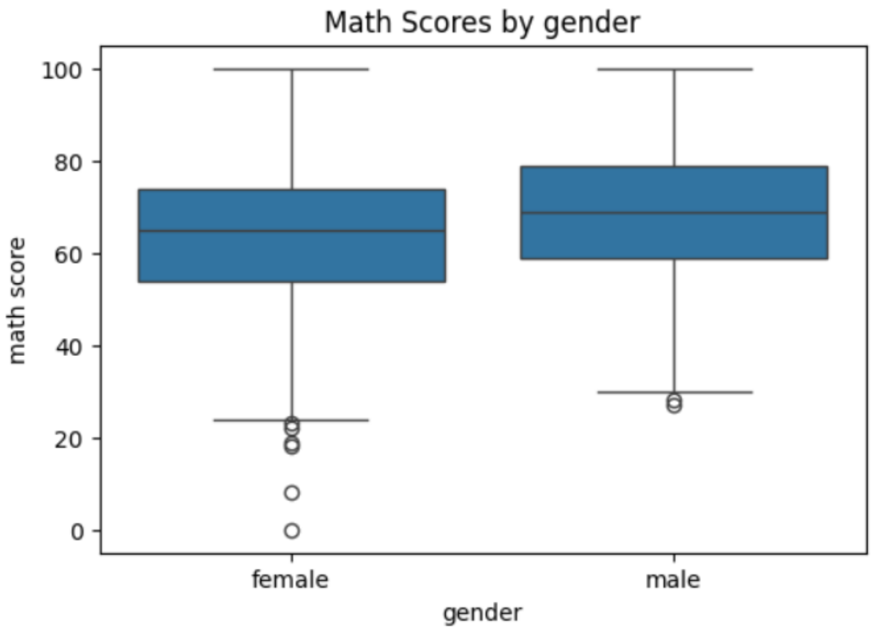
VI. Data visualization

Graph 1 : Math score Distribution



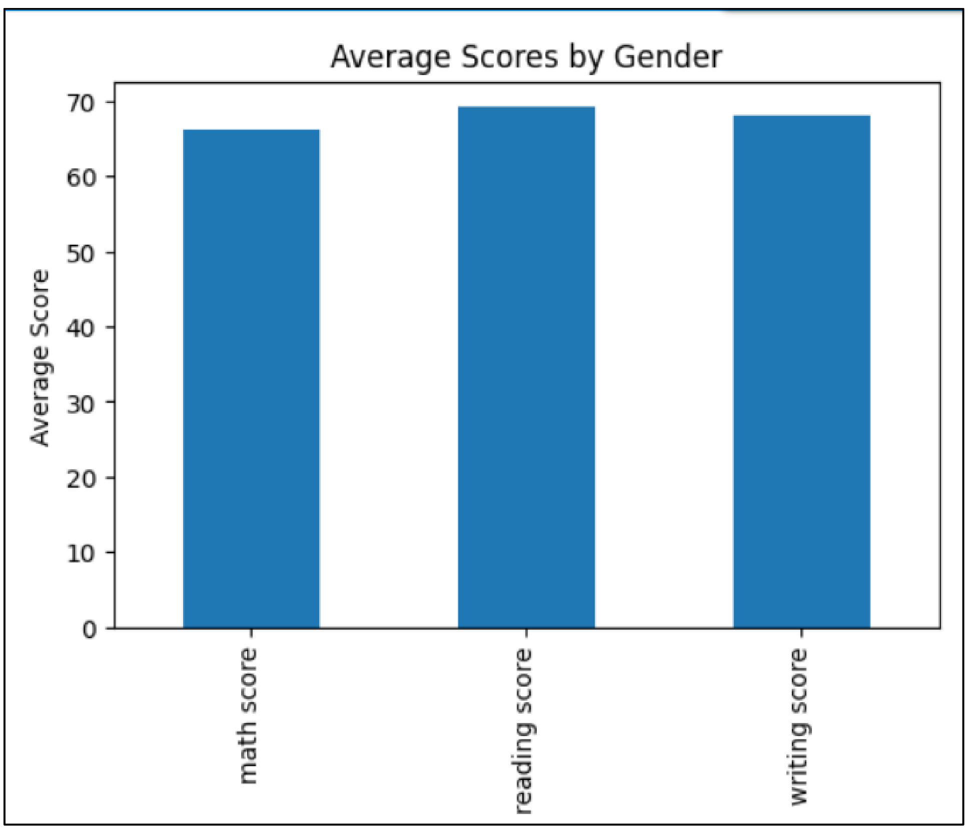
Observation: Most of the students scored between 60-80 marks.

Graph 2: Math Scores by Gender



Observation: Male and female students showed similar performance.

Graph 3: Average Subject Scores



Observation: Reading and writing scores are slightly higher than the maths scores.

VII. Conclusion:

The project successfully performed data cleaning and visualization on the Student Performance Dataset.The data was cleaned using pandas and the visualization was using Matplotlib and seaborn.The graphs helped us in understanding Student performance and demonstrated the importance of data preprocessng and the visualization.

VIII. References

- Kaggle Student Performance Dataset
- Python Documentation
- Pandas Documentation
- Matplotlib Documentation